

Notes: Coibion, Gorodnichenko, and Kumar (AER, 2018)

How Do Firms Form Their Expectations? New Survey Evidence

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1 Big picture

- **Question.** How do firms form expectations about macroeconomic variables, especially inflation, and do those expectations affect firm decisions? The paper asks whether managers are well informed, how they update beliefs when given new information, and whether belief changes translate into planned real actions.
- **Setting.** The evidence comes from a large quantitative survey of New Zealand firms. Respondents are general managers rather than professional forecasters, consumers, or financial-market participants.
- **Why firms matter.** Firms set prices, wages, hiring, investment, and production. Monetary policy depends on how these decision makers perceive inflation, unemployment, GDP growth, and other macroeconomic conditions.
- **Main empirical surprise.** Managers display large disagreement about both future and recent macroeconomic conditions. The dispersion is especially large for aggregate inflation, despite New Zealand's long history of inflation targeting.
- **Past beliefs and future beliefs are linked.** Firms that believe recent inflation was high also tend to forecast high future inflation. This suggests that errors in expectations often originate in misinformation or inattention about current and recent economic conditions.
- **Rational inattention interpretation.** The cross-sectional patterns are consistent with incentives to acquire information. Firms facing more competition, firms close to a price change, and firms with steeper profit functions tend to be better informed.
- **Experimental evidence.** When firms are randomly given information about recent macroeconomic variables, professional forecasts, the central bank's inflation target, or other firms' beliefs, they revise their forecasts in the direction of the signal. The revision patterns are broadly consistent with Bayesian updating.
- **Persistence and policy relevance.** Some informational effects are short lived. Information about the Reserve Bank of New Zealand's inflation target strongly affects beliefs on impact, but much of the short-run effect fades after six months.
- **Real effects.** Firms that receive target information and lower their inflation expectations subsequently reduce planned employment and investment relative to their earlier plans and relative to a control group. Prices and wages respond much less.
- **Economic takeaway.** Firms are not simply miniature professional forecasters. Aggregate inflation is often not important enough for managers to monitor closely, so expectations are dispersed, biased, and inattentive. Policy communication can move expectations, but it must break through this inattention.

2 Incremental contribution of the paper

- **From household and professional-forecaster expectations to firm expectations.** Much of the earlier evidence on expectations uses households, professional forecasters, financial markets, or policymakers. This paper directly measures the macroeconomic beliefs of firm managers.

- **A quantitative firm survey.** Many business surveys ask qualitative questions, such as whether prices will rise or fall. This survey asks numerical forecasts and probability distributions, making it possible to measure means, dispersion, uncertainty, and forecast/backcast errors.
- **Beliefs about the past and present.** The paper does not only ask about future outcomes. It asks managers about recent inflation, GDP growth, unemployment, and industry price changes. This lets the authors identify inattention to recent macroeconomic developments.
- **Panel dimension.** Multiple waves allow the paper to follow firms over time and measure the persistence of inattention. The panel element also helps distinguish temporary measurement noise from persistent information frictions.
- **Firm-level determinants of inattention.** The survey collects firm characteristics, market structure, pricing plans, competition, margins, and respondent characteristics. This allows a direct test of rational inattention mechanisms.
- **Experimental contribution.** The paper embeds information treatments in the survey. Random information provision lets the authors observe causal updating of beliefs rather than only correlations.
- **Bayesian updating evidence.** Firms revise forecasts toward new information, and firms with higher prior uncertainty revise more strongly. This provides direct evidence that managers can update sensibly when relevant information is put in front of them.
- **Expectations-to-actions contribution.** The second experiment links experimentally induced belief changes to planned employment, investment, prices, wages, unit costs, and sales. This moves beyond measuring beliefs to testing whether expectations matter for firm behavior.
- **Information-value and tracking contribution.** The paper asks which macro variables managers rank as important, which variables they track, and how much they would pay for professional forecasts. This connects measured attention to firm incentives.
- **Policy contribution.** The paper shows why expectation management and forward guidance may face a practical limitation: many firms do not track inflation closely because they do not perceive it as important for business decisions.

3 Data and measurement

3.1 Survey design and sampling

- The survey covers New Zealand firms and is implemented in six waves between 2013:IV and 2016:IV.
- The first wave includes 3,144 firms. Follow-up waves include subsets of these firms, and later waves add new firms and some returning firms.
- The survey focuses on general managers because these people are likely to influence pricing, hiring, investment, and production decisions.
- The sampling frame is built from business directories and covers small, medium, and large firms from several broad sectors.

- The core sectors include manufacturing, retail and wholesale trade, construction and transportation, and professional and financial services.
- The survey excludes some sectors such as government, community service, agriculture, fishing, mining, energy, gas, and water because these sectors often have special regulation, a small number of firms, or many tiny firms.
- Sampling weights correct for discrepancies between the survey sample and the employment distribution of firms in New Zealand.

Interpretation: The sample is designed to measure the expectations of actual business decision makers rather than experts. This is exactly why the survey can differ sharply from professional forecasts.

3.2 Survey waves and key modules

- **Wave 1** is the largest wave and asks many questions about inflation expectations, price setting, price-review frequency, and the expected timing and size of the next price change.
- **Wave 2** expands macroeconomic questions and provides follow-up data on price changes.
- **Wave 3** collects more respondent characteristics, including age, income, education, and tenure.
- **Wave 4** implements the first information experiment, in which firms receive randomly assigned macroeconomic signals and then revise forecasts.
- **Waves 5 and 6** implement the second experiment, centered on information about the Reserve Bank of New Zealand inflation target and later follow-up.

Interpretation: The design combines cross-sectional survey measurement, panel follow-up, and randomized information provision. This combination is why the paper can study static dispersion, persistence, updating, and behavior.

3.3 Forecasts, backcasts, and nowcasts

- A *forecast* is a belief about a future variable, such as inflation over the next twelve months.
- A *backcast* is a belief about a recent past value, such as inflation over the previous twelve months.
- A *nowcast* is a belief about a current or contemporaneous macro variable, such as the current unemployment rate.
- The paper asks about aggregate inflation, industry-specific inflation, unemployment, GDP growth, and in some waves the exchange rate.
- The survey also elicits probability distributions for forecasts in some waves, allowing the authors to measure each firm's uncertainty.

Interpretation: Backcasts and nowcasts are central to the paper. If managers do not even know recent macro conditions, then their future forecasts may be dispersed because they begin from different perceived facts.

3.4 Quality control and response credibility

- The authors compare survey answers against outside records when possible, such as administrative records for firm age and online prices for firms' products.
- They check internal consistency across repeated questions and related questions. Firms that say they review prices often also report more frequent price changes.
- They use follow-up waves to test whether stated expectations predict subsequent actions. Figure 1 shows that firms' expected price-change timing and size predict actual price changes.
- Extremely inconsistent or outlier responses are removed.

Interpretation: The paper relies on survey answers, so data quality matters. The validation exercises show that managers' responses are not random noise; they contain information about later firm behavior.

3.5 Comparison groups

- Firm forecasts are compared with forecasts from the Reserve Bank of New Zealand, Consensus Economics professional forecasters, and households in the Reserve Bank of New Zealand survey.
- Household inflation forecasts are trimmed by the Reserve Bank of New Zealand. For comparability, the authors apply the same trimming to firms' inflation forecasts when comparing firms with households.
- Central bank forecasts and professional forecasts provide a benchmark for more informed expectations.

Interpretation: The comparison is not simply about whether firms are "right." It asks whether the agents setting prices and production have beliefs more like professional forecasters, households, or something else.

4 Models and empirical specifications

4.1 Beliefs about the past and forecasts of the future

The paper uses the following regression to study whether a firm's beliefs about recent conditions predict its future forecasts:

$$F_t^i x_{t+12} = \alpha + \beta B_t^i x_t + \delta_j + \varepsilon_{it}.$$

- $F_t^i x_{t+12}$ is firm i 's twelve-month-ahead forecast for macro variable x .
- $B_t^i x_t$ is the firm's backcast or nowcast of the recent or current value of x .
- δ_j is an industry or sub-industry fixed effect.
- The coefficient β measures how strongly perceived recent conditions are linked to expected future conditions.

Interpretation: If β is positive and large, differences in future beliefs are partly explained by differences in perceived current facts. This is exactly what imperfect-information models predict.

4.2 Measuring inattention

The paper measures inattention using absolute errors about recent conditions:

$$|x_t - B_t^i x_t|.$$

For inflation, this is the absolute gap between actual recent inflation and the inflation rate the manager believes occurred.

- Larger absolute errors indicate lower attention or worse information about the macro variable.
- The authors focus especially on inflation because inflation errors are large, asymmetric, and strongly tied to forecasts.
- Regressions explain inflation errors using firm characteristics, competition, pricing incentives, and manager characteristics.

Interpretation: This is a direct measure of information quality. Instead of inferring inattention indirectly from sluggish forecasts, the paper measures whether managers know recent facts.

4.3 Determinants of inattention

A simplified version of the inattention regression is

$$|\pi_{t,t-12} - B_t^i \pi_{t,t-12}| = \alpha + X_i' \gamma + Z_i' \theta + P_i' \rho + M_i' \eta + \delta_j + u_i,$$

where:

- X_i includes firm characteristics such as firm age, employment, labor share, and foreign trade share.
- Z_i includes competition and market-power variables such as the number of competitors, margins, and price relative to competitors.
- P_i includes price-setting variables such as past price changes, industry PPI inflation, expected size of price change, and duration until next price change.
- M_i includes manager characteristics such as age, education, tenure, and income.

Interpretation: Rational inattention predicts that information should be better when the payoff to information is higher. Competition, upcoming price changes, and steep profit functions should reduce errors.

4.4 Persistence of inattention

The panel survey allows the paper to estimate whether inattention persists over time:

$$|x_t - B_t^i x_t| = \alpha + \beta |x_{t-1} - B_{t-1}^i x_{t-1}| + \delta_j + \varepsilon_{it}.$$

- β measures persistence in errors.
- A high β means that firms with large errors in one wave tend to have large errors in later waves.
- The estimates can be mapped into the degree of information rigidity.

Interpretation: Persistent errors are more important than one-time mistakes. They imply that firms repeatedly operate with limited macroeconomic information.

4.5 Bayesian updating with information treatments

The first experiment gives firms a signal s and observes how their posterior forecast changes. If firm i has prior mean μ_i and prior precision τ_i , and receives a common signal s with precision ψ_s , Bayesian updating implies

$$p_i = \mu_i + \frac{\psi_s}{\psi_s + \tau_i}(s - \mu_i).$$

Equivalently,

$$p_i - \mu_i = \frac{\psi_s}{\psi_s + \tau_i}(s - \mu_i).$$

The empirical regression can be written as

$$p_i = c + \beta\mu_i + \varepsilon_i,$$

where p_i is the posterior and μ_i is the prior. The slope β indicates how much of the prior remains after the signal.

- A lower β means a stronger response to the signal.
- If the signal is very precise, the posterior moves strongly toward the signal.
- If the prior is very precise, the posterior remains closer to the prior.

Interpretation: Firms are inattentive, but not irrational in the narrow sense of ignoring information when it is supplied. They often update in the direction and magnitude predicted by Bayesian learning.

4.6 Uncertainty and the size of revisions

Bayesian updating also predicts that firms with greater prior uncertainty should revise more for a given signal-prior gap. The paper estimates regressions based on the scaled revision:

$$\frac{p_i - \mu_i}{s - \mu_i} = c + \beta\sigma_i + \varepsilon_i,$$

where σ_i is the standard deviation implied by the firm's reported probability distribution.

- A positive coefficient means more uncertain firms put more weight on the signal.
- This is especially useful because the survey elicits probability distributions, not only point forecasts.

Interpretation: The probability-distribution questions are not decorative. They allow the authors to test the internal logic of Bayesian updating.

5 Identification and empirical design

5.1 What the descriptive survey evidence identifies

- The cross-sectional survey evidence identifies facts about dispersion, bias, disagreement, and inattention among firms.
- It can show that firms with particular characteristics are more or less informed.
- It cannot by itself prove that competition or pricing schedules causally reduce inattention.
- The descriptive regressions are best read as testing whether the patterns line up with rational inattention incentives.

Interpretation: The survey regressions support rational inattention, but they are not a randomized design for every firm characteristic.

5.2 Why the experiments matter

- The first information experiment randomly assigns firms to receive different pieces of information.
- The second experiment randomly provides information about the RBNZ inflation target to a subset of firms.
- Random assignment lets the authors interpret belief revisions as causal responses to information.
- Comparing treated firms with a control group also helps identify whether belief changes affect planned actions.

Interpretation: The experiments are the strongest part of the paper’s causal evidence. They show that information moves expectations and that expectation changes can move real planned decisions.

5.3 Endogeneity and interpretation of actions

- Without an experiment, firms with high inflation expectations may differ from other firms in many unobserved ways, making it difficult to interpret the relationship between expectations and actions.
- The second experiment addresses this by changing some firms’ inflation expectations through information about the central bank target.
- The paper compares firms whose priors are far from the target with firms whose priors are close to the target.
- The most informative group is firms with high priors about the RBNZ target: they receive information that leads them to lower inflation expectations.

Interpretation: The action results are not based on a simple correlation between expectations and investment. They exploit an induced belief change.

5.4 Limitations of identification

- The survey is from New Zealand, so external validity to larger economies or high-inflation countries must be considered carefully.
- The sample consists of firms willing to participate in the survey; nonresponse could matter.
- Some outcomes are planned actions rather than realized future actions.
- The information treatments are survey-based and may differ from real-world central bank communication campaigns.
- The paper measures macroeconomic attention and expectations, but not every managerial belief that could affect decisions.

Interpretation: The design is unusually rich, but not a complete solution to all issues. The safest conclusion is that firm expectations are measurable, dispersed, experimentally movable, and relevant for decisions in this setting.

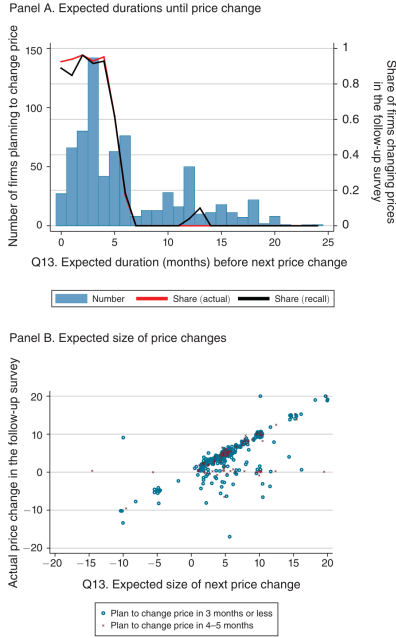
6 Tables and figures

6.1 Figure 1: Firms' Expected and Actual Price Changes; Table 1: Macroeconomic Forecasts of Firms and Other Economic Agents

Read:

- Figure 1 is a data-quality check. Panel A compares firms' expected duration until the next price change with the share that actually changed prices by the follow-up survey.
- Panel B compares the expected size of the next price change with the actual subsequent price change.
- The figure shows that survey expectations about firm-level prices are predictive of later firm behavior.
- Table 1 compares firms' twelve-month-ahead macro forecasts with the central bank, professional forecasters, and households across waves.
- Firms' inflation forecasts are much higher and more dispersed than those of professional forecasters, and they resemble households more than experts.
- For unemployment and GDP growth, the average firm forecast is closer to professional forecasts, but dispersion remains higher.

Economic message: Figure 1 establishes credibility of the survey responses. Table 1 establishes the main puzzle: managers, especially on inflation, do not look like professional forecasters even though they make business decisions affected by inflation.



Notes: In Panel A, the bars (left axis) show the number of firms reporting a given expected duration before next price change in the first wave. The lines show the fraction of firms that actually adjusted their prices between the first and second waves, grouped by each duration. The red line measures changes in prices as the difference in current prices reported in the two waves. The black line measures changes in prices as the change between the current price reported in the second wave and the previous price reported in the second wave. The previous price is the price three months ago for firms surveyed in December 2013 or January 2014 and six months ago for firms surveyed in September 2013, October 2013, or November 2013. Panel B plots firms' expectation of the size of their next price change (in percent) as reported in the first wave (x-axis) versus firms' actual percentage change in price between the first and second waves (y-axis) for firms that reported that they expected to change prices within the next five months. Circles and crosses indicate the expected duration (reported in the first wave) before the next price change. See Section IIA for details.

Figure 1: Firms' Expected and Actual Price Changes

TABLE 1—MACROECONOMIC FORECASTS OF FIRMS AND OTHER ECONOMIC AGENTS

	Central bank	Professional forecasters		Households		Firms	
	(1)	Mean (2)	SD (3)	Mean (4)	SD (5)	Mean (6)	SD (7)
<i>Panel A. 2013:IV (wave 1, number of observations: 3,144)</i>							
Inflation	1.3	2.0	0.2	3.6	2.4	5.3	3.2
<i>Panel B. 2014:I (wave 2, number of observations: 712)</i>							
Inflation	1.9	2.0	0.3	3.7	2.1	6.1	2.7
Unemployment	4.9	5.3	0.3	NA	NA	5.2	0.7
GDP growth	3.5	3.4	0.5	NA	NA	3.1	0.7
<i>Panel C. 2014:III (wave 3, number of observations: 1,601)</i>							
Inflation	1.6	1.9	0.2	3.5	2.4	4.1	2.5
<i>Panel D. 2014:IV (wave 4, number of observations: 1,257)</i>							
Inflation	1.1	1.7	0.3	3.1	2.0	4.5	2.8
Unemployment	5.2	5.2	0.3	NA	NA	5.9	1.2
GDP growth	3.5	3.0	0.3	NA	NA	3.6	1.0
<i>Panel E. 2016:II (wave 5, number of observations: 2,040)</i>							
Inflation	1.6	1.3	0.2	2.3	2.1	2.8	2.3
Unemployment	5.2	5.5	0.2	NA	NA	5.5	0.6
GDP growth	3.4	2.6	0.3	NA	NA	2.7	0.5
<i>Panel F. 2016:IV (wave 6, number of observations: 1,404)</i>							
Inflation	1.7	1.6	0.2	2.8	2.6	2.7	2.4
Unemployment	4.7	4.8	0.3	NA	NA	5.5	0.6
GDP growth	3.4	3.0	0.4	NA	NA	2.4	0.6

Notes: The table reports recent 12-month-ahead forecasts and the dispersion in forecasts of different macroeconomic variables for different agents. Inflation refers to annual changes in prices (CPI when specified), unemployment refers to the unemployment rate at a specific quarter, GDP growth refers to annual changes in real GDP. Central bank forecasts are from Monetary Policy Statements of the Reserve Bank of New Zealand. Professional forecasts are from Consensus Economics. Households are from the Reserve Bank of New Zealand's Survey of Households. The inflation forecasts of households are trimmed by the Reserve Bank of New Zealand and exclude all forecasts of inflation above 15 and below -2 percent, so same trimming is applied to firms' inflation forecasts for comparison. Other firm forecasts are unadjusted. Moments for firms and households are calculated using sampling weights. See Section IIA for details.

Table 1: Macroeconomic Forecasts of Firms and Other Economic Agents

6.2 Figure 2: Inflation in New Zealand; Table 2: Recent Values and Nowcasts of Firms and Other Economic Agents

Read:

- Figure 2 places survey inflation expectations in the history of New Zealand inflation.
- Panel A shows that actual CPI inflation has been low and stable since the adoption of inflation targeting, while survey expectations are often much higher.
- Panel B shows that movements in firms' inflation expectations comove with petrol-price movements, suggesting that salient price changes influence aggregate inflation beliefs.
- Table 2 compares beliefs about recent macroeconomic values. Firms are not only dispersed about the future; they are also dispersed about recent inflation and unemployment.
- Firms' nowcasts/backcasts of inflation are much higher and more dispersed than recent official data.

Economic message: The figure and table show that the issue is not only forecasting difficulty. Many firms appear misinformed about recent macroeconomic facts, and salient prices such as petrol may shape their inflation perceptions.

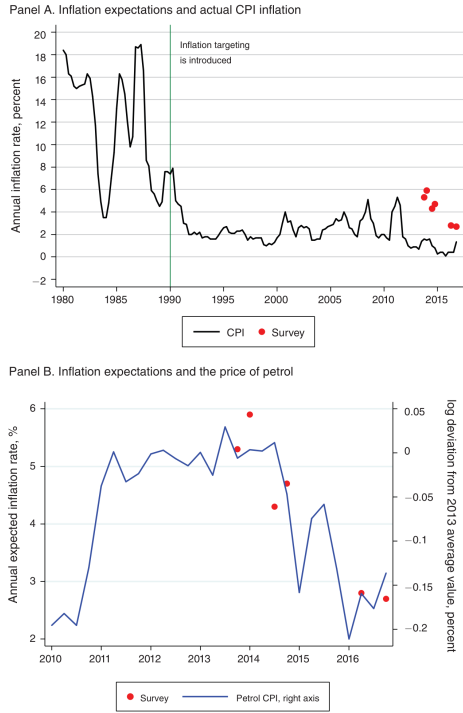


FIGURE 2. INFLATION IN NEW ZEALAND

Notes: The figure plots annual CPI inflation in New Zealand, price for petrol products (measured as log percent deviation from the average price in 2013), and mean inflation expectations reported in six survey waves.

Figure 2: Inflation in New Zealand

TABLE 2—RECENT VALUES AND NOWCASTS OF FIRMS AND OTHER ECONOMIC AGENTS

	Recent data (1)	Households		Firms	
		Mean (2)	SD (3)	Mean (4)	SD (5)
<i>Panel A. 2013:IV (wave 1, number of observations: 3,144)</i>					
Inflation	1.4	3.1	2.0	4.4	3.5
<i>Panel B. 2014:I (wave 2, number of observations: 712)</i>					
Inflation	1.6	2.9	1.8	5.5	3.3
Unemployment	5.6	NA	NA	6.5	1.4
<i>Panel C. 2014:III (wave 3, number of observations: 1,601)</i>					
Inflation	1.6	2.9	2.0	NA	NA
<i>Panel D. 2014:IV (wave 4, number of observations: 1,257)</i>					
Inflation	1.0	2.9	2.2	3.9	2.4
Unemployment	5.2	NA	NA	6.1	1.2
GDP growth	3.4	NA	NA	3.7	1.2
<i>Panel E. 2016:II (wave 5, number of observations: 2,040)</i>					
Inflation	0.4	1.8	1.5	2.6	2.1
<i>Panel F. 2016:IV (wave 6, number of observations: 1,404)</i>					
Inflation	0.4	2.4	2.4	NA	NA

Notes: The table reports recent values and nowcasts/backcasts as well as the dispersion in nowcasts of different macroeconomic variables for different agents. Inflation refers to annual changes in prices (CPI when specified) over the past 12 months, unemployment refers to the unemployment rate in that quarter, GDP growth refers to annual changes in real GDP. Households are from the Reserve Bank of New Zealand's Survey of Households. The inflation nowcasts of households are trimmed by the Reserve Bank of New Zealand and exclude all nowcasts of inflation above 15 and below -2 percent, so same trimming is applied to firms' inflation nowcasts for comparison. Other firm nowcasts are unadjusted. Moments for firms and households are calculated using sampling weights. See Section IIA for details.

Table 2: Recent Values and Nowcasts of Firms and Other Economic Agents

6.3 Table 3: Beliefs about Future and Past Values of Macroeconomic Variables; Figure 3: Distribution of Inflation Expectations across Firms

Read:

- Table 3 regresses future forecasts on backcasts or nowcasts. For every macro variable, beliefs about the recent past strongly predict future forecasts.
- The inflation coefficient is positive and economically meaningful. Managers who think past inflation was high are likely to forecast high future inflation.
- Including firm fixed effects does not eliminate the link for variables where the specification is feasible.
- Figure 3 splits firms into “informed” and “uninformed” groups based on the size of their backcast errors about recent inflation.
- In wave 1, informed firms have a much tighter and lower inflation forecast distribution. Uninformed firms have a long right tail of very high inflation forecasts.
- By wave 4, the distinction based on wave-1 information becomes less stable, showing that being informed is not a permanent firm type.

Economic message: Differences in expectations are not just differences in tastes or optimism. They are tied to different perceived facts. When managers start from wrong inflation backcasts,

their future inflation forecasts are also distorted.

	No firm fixed effects (1)	Firm fixed effects (2)
<i>Inflation rate: aggregate</i>	0.339 (0.025)	0.286 (0.044)
Observations	5,130	3,531
R^2	0.389	0.730
<i>Inflation rate: industry</i>	1.038 (0.014)	
Observations	1,154	
R^2	0.959	
<i>Unemployment rate</i>	0.863 (0.012)	0.758 (0.091)
Observations	1,842	770
R^2	0.826	0.828
<i>GDP growth rate</i>	0.909 (0.010)	
Observations	1,194	
R^2	0.928	
<i>Exchange rate</i>	0.998 (0.002)	
Observations	1,035	
R^2	0.994	

Notes: The table reports Huber-robust regressions of firms' 12-month-ahead forecasts of a given variable, indicated in the left column, on the backcast (over previous 12 months) or nowcast of the variable. Nowcasts are used for the unemployment rate and the exchange rate. Sub-industry fixed effects are included but not reported. In column 1, a constant term is included but not reported. Column 2 reports results for the specification when data are pooled across waves and firm fixed effects are included. Aggregate inflation uses data from waves 1, 2, 4, and 5. Unemployment rate uses data from waves 1 and 2. Industry inflation rate, GDP growth rate, and Exchange rate use data from wave 4. Robust standard errors (clustered at the 3-digit ANZ SIC level) are reported in parentheses. See Section IIIA for details.

The specific variables for which we have at least one set of forecasts and nowcasts/

Table 3: Beliefs about Future and Past Values of Macroeconomic Variables

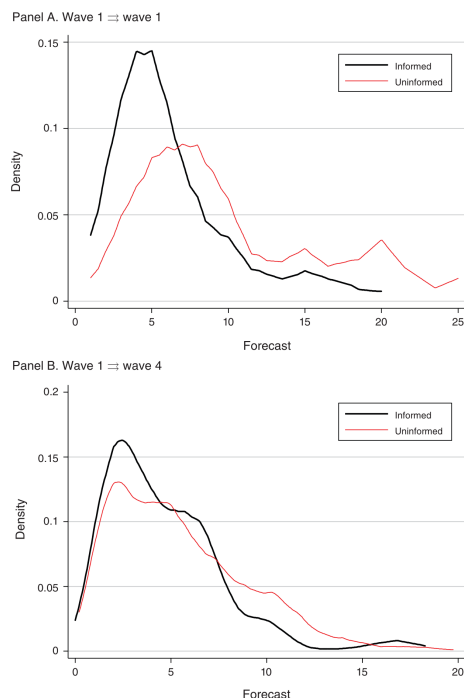


FIGURE 3. DISTRIBUTION OF INFLATION EXPECTATIONS ACROSS FIRMS

Notes: The two panels plot distributions of firms' inflation forecasts over the next 12 months. Panel A plots the distribution for "informed" firms (those with errors about recent inflation were less than 2 percentage points) versus "uninformed" firms in the first wave. Panel B plots equivalent distributions in the fourth wave of the survey, but continuing to identify "informed" and "uninformed" firms based on the size of their backcast errors in the first wave.

Figure 3: Distribution of Inflation Expectations across Firms

6.4 Figure 4: Distributions of Errors about Recent Macroeconomic Conditions; Table 4: Determinants of Firm Inattention

Read:

- Figure 4 plots distributions of errors about recent macroeconomic conditions.
- Panel A shows that aggregate inflation errors are unusually dispersed and asymmetric. Many firms believe recent inflation was much higher than it actually was.
- Errors about industry-specific inflation are much more symmetric and centered, suggesting that firms know their local price environment better than aggregate inflation.
- Panel B shows that inflation errors differ strongly by industry. Manufacturing and trade firms are more informed than professional/financial services and construction/transportation firms.
- Table 4 explains absolute inflation errors using firm, competition, price-setting, and manager variables.
- Firms with fewer competitors, longer expected time until price change, flatter profit functions, and more foreign exposure tend to make larger errors.
- Manager characteristics add relatively little once firm characteristics are included.

Economic message: The evidence supports rational inattention. Managers pay attention when attention is valuable for business decisions. Aggregate inflation is often not monitored closely, while industry and firm-specific prices are better understood.

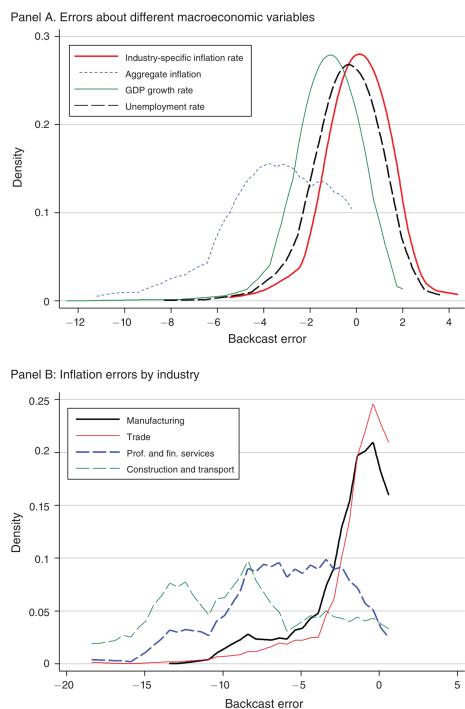


FIGURE 4. DISTRIBUTIONS OF ERRORS ABOUT RECENT MACROECONOMIC CONDITIONS

Notes: Panel A plots kernel density estimates of distributions of errors about recent values of different macroeconomic variables made by firms. Panel B plots kernel density estimates of the distribution of errors about recent inflation made across different industries. Sample weights are used for both panels. See Section IIIB for details.

Figure 4: Distributions of Errors about Recent Macroeconomic Conditions

	(1)	(2)	(3)
Firm characteristics			
log(age)	0.114 (0.031)		0.175 (0.059)
log(employment)	0.384 (0.061)		0.486 (0.074)
Labor's share of costs	-0.005 (0.004)		-0.005 (0.005)
Foreign trade share	0.009 (0.003)		0.011 (0.004)
Number of competitors	-0.006 (0.002)		-0.005 (0.003)
Average margin	0.002 (0.004)		0.003 (0.003)
Price relative to competitors	0.003 (0.003)		0.001 (0.003)
Firm's past price changes	-1.165 (0.264)		-0.872 (0.364)
Industry PPI inflation	-0.011 (0.007)		-0.015 (0.007)
Expected size of price change	-0.002 (0.003)		-0.001 (0.004)
Duration until price change	0.028 (0.009)		0.029 (0.013)
Absolute slope of profit function	-0.203 (0.044)		-0.323 (0.059)
Manager characteristics			
Age		-0.002 (0.003)	-0.003 (0.004)
Education:			
Some college		0.032 (0.069)	0.073 (0.066)
College		0.064 (0.069)	0.086 (0.084)
Graduate (MA+)		-0.058 (0.078)	-0.065 (0.092)
Tenure		0.029 (0.006)	-0.013 (0.008)
Income		0.000 (0.001)	-0.000 (0.001)
Industry fixed effects	Yes	Yes	Yes
Observations	2,912	1,332	1,338
R ²	0.799	0.828	0.834

Notes: The table reports results for the Huber robust regression. Dependent variable is the absolute value of firm errors about past 12-month inflation from wave 1 survey. Omitted category for manager's education is "high school diploma or less." Sample weights are applied to all specifications. Robust standard errors (clustered at the 3-digit ANZ SIC level) are reported in parentheses. See Section IIIC for details.

Table 4: Determinants of Firm Inattention

6.5 Figure 5: Inflation Backcasts and Forecasts by Firm Characteristic; Table 5: Persistence of Inattention

Read:

- Figure 5 groups firms by time until next price adjustment, number of competitors, and slope of the profit function.
- Firms expecting to adjust prices sooner have lower backcasts and lower forecasts of inflation.
- Firms with more competitors and steeper profit functions also have lower inflation backcasts and forecasts.
- This visualizes the same rational-inattention logic as Table 4: stronger incentives to process information are associated with better macro beliefs.
- Table 5 estimates the persistence of absolute errors across waves.

- Inflation errors are highly persistent between wave 1 and wave 2 and remain meaningfully persistent between wave 1 and wave 4.
- Errors for output gap and unemployment also show strong persistence.

Economic message: The key message is that inattention is systematic and durable. It is not merely a one-time slip or survey confusion; firms with poor information today are likely to have poor information later.

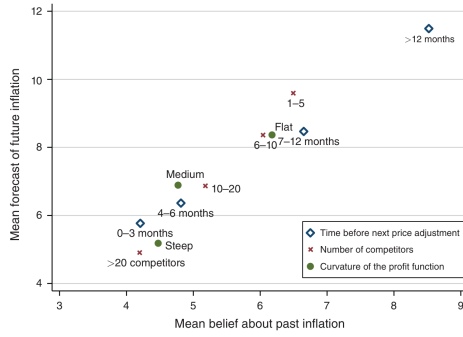


FIGURE 5. INFLATION BACKCASTS AND FORECASTS BY FIRM CHARACTERISTIC

Notes: The figure plots mean backcasts and forecasts of inflation for firms grouped by firm characteristics in the first wave. One grouping is by number of months until next expected price change, a second grouping is by the number of competitors, and the third grouping is by the tercile of the distribution of the steepness of the profit function. See Section IIIC for details.

Figure 5: Inflation Backcasts and Forecasts by Firm Characteristic

TABLE 5—PERSISTENCE OF INATTENTION	
Absolute error in the follow-up surveys	
<i>Panel A. Inflation over the previous 12 months, wave 2</i>	
Absolute error for inflation in wave 1	0.744 (0.026)
Observations	667
R^2	0.719
<i>Panel B. Inflation over the previous 12 months, wave 4</i>	
Absolute error for inflation in wave 1	0.410 (0.018)
Observations	1,211
R^2	0.554
<i>Panel C. Output gap, wave 2</i>	
Absolute error for output gap in wave 1	0.994 (0.004)
Observations	569
R^2	0.981
<i>Panel D. Unemployment rate, wave 4</i>	
Absolute error for unemployment rate in wave 2	0.771 (0.026)
Observations	441
R^2	0.794

Notes: The table reports Huber-robust regressions of firms' absolute errors for inflation over the last 12 months (panels A and B), the contemporaneous output gap (panel C), and contemporaneous unemployment rate (panel D) in waves 1, 2, and 4 on firms' errors over the same variables in waves 1 and 2. Sample weights are used in all specifications. Constant is included but not reported. Robust standard errors (clustered at the 3-digit ANZ SIC level) are reported in parentheses. See Section IIIC for details.

Table 5: Persistence of Inattention

6.6 Figure 6: Forecast Updating by Firms upon Receiving New Information; Table 6: Information Updates

Read:

- Figure 6 plots priors against posteriors after firms receive new information.
- Forecasts move toward the signal. The fitted lines have slopes less than one, which is consistent with partial updating toward new information.
- The updating response is strongest for inflation. In Table 6, the pooled inflation slope on the prior is 0.339, lower than the corresponding slopes for unemployment and GDP growth.
- Inflation forecasts respond particularly strongly to information about the central bank target and recent inflation.
- Panel B of Table 6 shows that prior uncertainty predicts larger revisions for most inflation information treatments.

Economic message: Firms are inattentive before receiving information, but they are not incapable of updating. When relevant information is made salient, firms revise beliefs in a way broadly consistent with Bayesian learning.

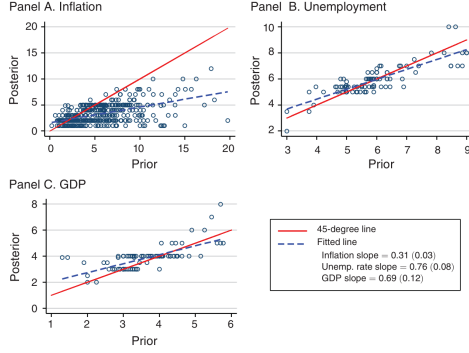


FIGURE 6. FORECAST UPDATING BY FIRMS UPON RECEIVING NEW INFORMATION

Notes: Each panel plots initial forecasts of firms in wave 4 (prior) against their revised forecasts (posterior) after being provided with new information. Panels include the 45-degree line and regression lines. Each panel is for forecasts of individual macroeconomic variable. See Section IV for details.

Figure 6: Forecast Updating by Firms upon Receiving New Information

Information source:	Inflation						Unempl.	GDP
	Pool (1)	SPF (2)	CB target (3)	CB target + SPF (4)	π_{t-1} (5)	$E_{t-1}\pi_{t+12}$ (6)	UE_{t-12} (7)	GDP_{t-12} (8)
Panel A. Dependent variable: Posterior p_t								
Prior, μ_t	0.339 (0.021)	0.309 (0.062)	0.228 (0.047)	0.284 (0.038)	0.282 (0.041)	0.567 (0.033)	0.647 (0.042)	0.565 (0.067)
Observations	500	100	100	100	100	100	98	99
R^2	0.375	0.319	0.195	0.273	0.347	0.676	0.737	0.498
Panel B. Dependent variable: Scaled revision of posterior: $\frac{p_t - \mu_t}{s - \mu_t}$								
Uncertainty, σ_t	0.058 (0.012)	0.065 (0.036)	-0.024 (0.023)	0.041 (0.021)	0.036 (0.013)	0.140 (0.027)	0.294 (0.136)	-0.471 (0.178)
Observations	448	86	78	91	89	98	81	80
R^2	0.082	0.021	0.011	0.028	0.029	0.118	0.040	0.055

Notes: Panel A reports results for specification (4) where the dependent variable is the posterior point prediction of the variable indicated in the first row of the table and the regressor is the prior, i.e., the point prediction implied by the reported probability distribution for the corresponding variable. The prior is the belief of a firm before the firm is presented with additional information. The posterior is the belief of a firm after the firm is presented with additional information. Fixed effects for source of information are included in column 1 but not reported. Panel B reports result for specification (6) where the dependent variable is the revision in beliefs (posterior minus prior) scaled by the difference between the signal s and the prior for the variable indicated in the first row of the table. The posterior and prior are defined as in panel A. The regressor is the standard deviation implied by the probability distribution for the corresponding variable. Fixed effects for source of information are included in column 1 but not reported. To minimize the effects of extreme observations, the sample in each column is constrained to include only observations with $\left|\frac{p_t - \mu_t}{s - \mu_t}\right| \leq 2$. Huber robust regression is used for all specifications. Robust standard errors are reported in parentheses. See Section IV for more details.

Table 6: Information Updates

6.7 Table 7: Effects of Information about RBNZ Inflation Target on Firms' Beliefs and Actions; Table 8: Ranking of Attention and Associated Backcast Errors and Forecast Uncertainty

Read:

- Table 7 studies the second experiment, where firms receive information about the Reserve Bank of New Zealand's inflation target.
- Firms with high prior beliefs about the target reduce their one-year and five-year inflation forecasts immediately after the information treatment.
- After six months, the effect on short-run inflation expectations largely disappears, while long-run expectations remain somewhat lower.
- The same firms reduce planned employment growth and investment relative to initial plans and relative to the control group.
- Prices and wages show little average response.
- Table 8 links attention rankings to willingness to pay, forecast uncertainty, and backcast errors.
- When firms rank one macro variable as more important, they are willing to pay more for professional forecasts of that variable, have lower forecast uncertainty, and make smaller backcast errors for it.

Economic message: Table 7 shows that expectation shocks can affect real plans, especially employment and investment. Table 8 shows why inflation is often neglected: firms track and value information more when it matters for their own decisions.

TABLE 7—EFFECTS OF INFORMATION ABOUT RBNZ INFLATION TARGET ON FIRMS’ BELIEFS AND ACTIONS

	No controls for firm characteristics		Controls for firm characteristics	
	High prior about RBNZ target (4 or more percent)	Low prior about RBNZ target (1 to 3 percent)	High prior about RBNZ target (4 or more percent)	Low prior about RBNZ target (1 to 3 percent)
	(1)	(2)	(3)	(4)
<i>Forecast error in firm-level outcome</i>				
Sales growth	0.044 (0.301)	0.202 (0.336)	0.032 (0.306)	0.227 (0.350)
Wage growth	-0.112 (0.069)	-0.037 (0.055)	-0.121 (0.072)	-0.046 (0.055)
Unit cost growth	-0.069 (0.107)	0.287 (0.168)	-0.142 (0.112)	0.285 (0.161)
Investment	-1.954 (1.075)	0.222 (0.582)	-2.129 (1.125)	0.227 (0.597)
Employment growth	-2.925 (0.949)	-0.001 (0.411)	-3.544 (1.007)	-0.043 (0.552)
Price change	-0.103 (0.077)	0.025 (0.116)	-0.095 (0.077)	0.031 (0.117)
<i>Change in expectations immediately after treatment</i>				
One-year-ahead inflation	-1.271 (0.220)	0.475 (0.162)	-1.212 (0.208)	0.467 (0.151)
Five-year-ahead inflation	-0.892 (0.182)	-0.108 (0.161)	-0.960 (0.202)	-0.147 (0.162)
One-year-ahead unemployment	0.046 (0.116)	-0.173 (0.133)	0.100 (0.127)	-0.190 (0.130)
One-year-ahead GDP growth rate	0.176 (0.077)	0.217 (0.119)	0.189 (0.076)	0.214 (0.116)
<i>Change in expectations six months after treatment</i>				
One-year-ahead inflation	0.310 (0.157)	0.024 (0.096)	0.341 (0.161)	-0.031 (0.092)
Five-year-ahead inflation	-0.405 (0.202)	-0.194 (0.176)	-0.389 (0.193)	-0.225 (0.175)
One-year-ahead unemployment	-0.186 (0.096)	-0.091 (0.148)	-0.157 (0.090)	-0.090 (0.146)
One-year-ahead GDP growth rate	0.043 (0.063)	0.146 (0.105)	0.053 (0.058)	0.115 (0.094)
RBNZ inflation target	0.073 (0.150)	-0.026 (0.089)	0.073 (0.142)	0.015 (0.082)

Notes: The table shows estimates of the treatment effect of providing information about the inflation target of the Reserve Bank of New Zealand on firms with priors close to the true target (columns 2 and 4) and on firms with priors far from the true target (columns 1 and 3). Influential observations are identified as observations that move the estimation by more than 0.5 of the standard error. These observations are excluded. Robust standard errors are reported in parentheses. See Section IV for more details.

TABLE 8—RANKING OF ATTENTION AND ASSOCIATED BACKCAST ERRORS AND FORECAST UNCERTAINTY

X Y	Regressor: $1(Rank_i^X > Rank_i^Y)$			Regressor: $Rank_i^X - Rank_i^Y$		
	Inflation UE (1)	Inflation GDP (2)	UE GDP (3)	Inflation UE (4)	Inflation GDP (5)	UE GDP (6)
<i>Panel A. Dependent variable is the relative size of willingness to pay for a professional forecast $\log\left(\frac{Pay_i^X}{Pay_i^Y}\right)$</i>						
Rank regressor	0.760 (0.012)	0.831 (0.015)	0.607 (0.011)	0.286 (0.003)	0.293 (0.003)	0.271 (0.004)
Observations	1,212	1,217	1,215	1,214	1,221	1,217
R ²	0.798	0.824	0.794	0.870	0.892	0.804
<i>Panel B. Dependent variable is the relative uncertainty in forecasts $\log\left(\frac{\sigma_i^X}{\sigma_i^Y}\right)$</i>						
Rank regressor	-0.183 (0.025)	-0.107 (0.032)	-0.050 (0.027)	-0.068 (0.008)	-0.028 (0.011)	-0.010 (0.011)
Observations	1,067	1,108	1,052	1,067	1,109	1,053
R ²	0.053	0.017	0.004	0.062	0.011	0.001
<i>Panel C. Dependent variable is the relative size of backcast errors $\log\left(\frac{ B_t X_{t-h} - X_{t-h} }{ B_t Y_{t-h} - Y_{t-h} }\right)$</i>						
Rank regressor	-2.001 (0.059)	-1.724 (0.043)	-0.929 (0.053)	-0.723 (0.017)	-0.530 (0.014)	-0.471 (0.020)
Observations	1,213	1,199	1,198	1,217	1,188	1,198
R ²	0.650	0.601	0.263	0.678	0.625	0.352

Notes: Panel A: $\log\left(\frac{Pay_i^X}{Pay_i^Y}\right)$ is the log of the ratio of willingness to pay (\$/year) for a professional forecast for variable X to willingness to pay (\$/year) for a professional forecast for variable Y. Panel B: $\log\left(\frac{\sigma_i^X}{\sigma_i^Y}\right)$ is the relative uncertainty in forecasts where σ_i^X measures uncertainty in forecasts from the probability distribution for variable X. Panel C: relative size of backcast errors $\log\left(\frac{|B_t X_{t-h} - X_{t-h}|}{|B_t Y_{t-h} - Y_{t-h}|}\right)$ where $B_t X_{t-h}$ is the backcast made at time t for variable X at time t - h. The horizon h is 12 months for inflation and GDP growth rate and 0 for the unemployment rate.

We use this question to rank variables in terms of relative attention:

“Which macroeconomic variables are most important to you in making your business decisions? Please rank the variables below from 1 (most important) to 3 (least important)”

- Unemployment rate
- GDP
- Inflation
- None of these is important to my decisions

$Rank_i^X - Rank_i^Y$ is the difference in ranks of variables X and Y as perceived by firm i. Ranks can take values 1, 2, 3. Thus the maximum difference is 2 and the minimum is -2. A higher value of the difference indicates that variable X is more important than variable Y. $1(Rank_i^X > Rank_i^Y)$ is the dummy variable equal to 1 if firm i thinks that variable X is more important for firms business decisions than variable Y. All estimates are based on Huber robust regressions. Sample weights are applied in all specifications. Robust standard errors (clustered at the 3-digit ANZ SIC level) are reported in parentheses. See Section V for details.

Table 7: Effects of Information about RBNZ Inflation Target on Firms’ Beliefs and Actions

Table 8: Ranking of Attention and Associated Backcast Errors and Forecast Uncertainty

6.8 Table 9: Macroeconomic Variables: Importance for Business Decisions and Tracking; Table 10: Complementarity in Acquisition of Information

Read:

- Table 9 reports how firms rank inflation, unemployment, and GDP growth for business decisions and whether they track each variable.
- Inflation is often not tracked. Less than half of firms report tracking inflation, while a much larger share tracks GDP.
- Firms that do not track inflation make substantially larger inflation backcast errors and have higher inflation forecasts.
- The table also reports willingness to pay for forecasts. Firms value forecasts more for variables they rank as important.
- Table 10 studies strategic complementarity in information acquisition and price setting.
- Firms with stronger price complementarity are more likely to prefer public signals and more likely to wait for other firms to change prices first.
- These patterns support models in which strategic complementarities make information about others’ beliefs and actions especially valuable.

Economic message: The final tables connect individual inattention to economic incentives. Firms choose what to follow. Inflation is neglected when it is not perceived as central to business decisions, while coordination motives make public information and competitors’ price changes valuable.

TABLE 9—MACROECONOMIC VARIABLES: IMPORTANCE FOR BUSINESS DECISIONS AND TRACKING

Importance for business decisions (1 = high, 3 = low)	Follow (percent) (1)	Do not follow (percent) (2)	Willingness to pay for a professional forecast (\$/year) (3)
<i>Panel A. Inflation</i>			
Shares, percent			
1	41.87	0.10	211.38
2	3.63	9.47	138.85
3	1.57	43.36	92.04
Total	47.07	52.93	148.26
Backcast error	1.10	4.96	
Forecast	3.34	5.85	
Forecast uncertainty (SD)	1.75	2.12	
<i>Panel B. Unemployment</i>			
Shares, percent			
1	18.02	1.13	166.60
2	39.32	8.83	115.88
3	13.99	18.71	110.73
Total	71.33	28.67	123.91
Backcast error	0.46	1.98	
Forecast	5.59	6.80	
Forecast uncertainty (SD)	0.79	0.71	
<i>Panel C. GDP</i>			
Shares, percent			
1	37.01	1.87	168.32
2	30.54	8.21	132.94
3	11.43	10.94	102.69
Total	78.99	21.01	139.93
Backcast error	1.02	2.42	
Forecast	3.50	4.15	
Forecast uncertainty (SD)	0.73	0.73	

Notes: The table shows shares of firms reporting importance of a given macroeconomic variable for their business decisions and whether they track the variable. The differences between “follow” and “do not follow” means for forecasts, backcast errors, and forecast uncertainty are statistically different from zero at 1 percent for all cases but one: forecast uncertainty for GDP. Column 3 shows the average willingness to pay for a professional forecast of a given variable. See Section V for details.

The mapping between firms’ answers as to which variables they track and the relative importance of different macroeconomic variables is summarized in

Table 9: Macroeconomic Variables: Importance for Business Decisions and Tracking

TABLE 10—COMPLEMENTARITY IN ACQUISITION OF INFORMATION

	No controls (1)	Sub-industry fixed effects (2)	Sub-industry fixed effects Firm controls Manager controls (3)
<i>Panel A. Information complementarity</i>			
Price complementarity	0.311 (0.036)	0.313 (0.037)	0.326 (0.034)
Observations	1,252	1,252	1,135
R ²	0.153	0.172	0.214
<i>Panel B. Importance of waiting for other firms</i>			
Price complementarity	0.613 (0.298)	0.618 (0.302)	0.828 (0.314)
Observations	1,241	1,242	1,125
R ²	0.004	0.019	0.077
<i>Panel C. Revision of inflation expectations when the main competitor raises its price</i>			
Importance of waiting for other firms	0.151 (0.028)	0.150 (0.028)	0.150 (0.030)
Observations	1,252	1,252	1,135
R ²	0.033	0.048	0.080

Notes: Panel A: The dependent variable is information complementarity which is a dummy variable equal to 1 if a firm picks “The source that can be seen by other firms” and 0 otherwise. The regressor is price complementarity which measures (in percent, absolute value) by how much sales of a given firm fall when a typical firm in your industry cuts its price by 10 percent. The response is divided by 10.

Panel B: The dependent variable is the dummy variable equal to 1 when a firm chooses “Wait until other firms make a price adjustment” in response to “Suppose you want to adjust your prices but you are uncertain about the state of the economy. What would you do?” and 0 otherwise. The response is divided by 10. Price complementarity is defined as in panel A. Influential observations are identified as observations that move the estimation by more than 0.2 of the standard error. These observations are excluded.

Panel C: The dependent variable is the response to the following question: “Suppose your main competitor raises the price of its product by 10 percent. By how much would you revise your expectation of inflation over the next 12 months.” The response is divided by 10. The regressor is the dummy variable equal to 1 when a firm chooses “Wait until other firms make a price adjustment” in response to “Suppose you want to adjust your prices but you are uncertain about the state of the economy. What would you do?” and 0 otherwise.

Robust standard errors (clustered at the 3-digit ANZ SIC level) are reported in parentheses. See Section V for details.

Table 10: Complementarity in Acquisition of Information

7 Main findings by theme

7.1 Firms disagree much more than experts

- Professional forecasters show very low disagreement about inflation in the New Zealand data.
- Firms show much more disagreement, similar in spirit to household surveys.
- The disagreement is most striking for aggregate inflation, not necessarily for every macro variable.
- The high dispersion exists even in a country with a long-standing inflation-targeting regime.

Interpretation: Inflation targeting may stabilize actual inflation without making all firms equally informed about inflation. The lack of volatility may even reduce the incentive to pay attention.

7.2 Aggregate inflation is special

- Firms understand their own prices and industry-specific price changes better than aggregate inflation.
- Errors about aggregate inflation are large and asymmetric; errors about industry-specific inflation are more centered.

- The asymmetry implies that many firms think inflation was much higher than it actually was.
- Petrol prices appear to be a salient influence on aggregate inflation expectations.

Interpretation: Managers may extrapolate from salient or frequently observed prices, or from their own business environment, when forming beliefs about aggregate inflation.

7.3 Inattention reflects incentives

- Firms with more competitors tend to have better inflation information.
- Firms closer to changing prices tend to have better information.
- Firms with steeper profit functions tend to have better information.
- Firms ranking a macro variable as important for decisions are more likely to track it and make smaller errors about it.

Interpretation: These facts fit rational inattention. Attention is costly, so firms allocate it toward variables with higher expected payoffs.

7.4 Firms update when information is supplied

- Firms revise forecasts in the direction of the signal they receive.
- They revise more when the prior-signal gap is larger.
- They revise more when prior uncertainty is larger.
- They respond especially to central bank target information and recent inflation information.

Interpretation: Inattention is not the same as irrational refusal to learn. Firms often update sensibly once information is delivered and noticed.

7.5 Expectation changes can affect decisions

- In the second experiment, firms with high prior beliefs about the RBNZ target lower inflation expectations after receiving target information.
- These firms reduce planned employment growth and investment relative to their previous plans and relative to controls.
- Prices and wages are relatively unchanged in the short horizon.

Interpretation: The link from expectations to firm behavior is real, but the margin may be quantities first rather than prices and wages immediately.

8 Policy implications

8.1 Forward guidance and communication

- If firms are inattentive to inflation, central bank communication may not automatically reach the agents whose decisions matter.
- Information about the inflation target can change beliefs on impact, but effects can decay quickly.

- Persistent communication may be required if policymakers want expectations to remain anchored at the firm level.

Interpretation: The paper does not say communication never works. It says that one-shot communication may not be enough when firms have little incentive to track inflation.

8.2 Anchoring of expectations

- Professional forecaster expectations may look anchored, but firm expectations can still be dispersed and biased.
- Aggregate statistics about average expectations can hide very wide firm-level heterogeneity.
- If firms' expectations are not anchored, monetary policy risks and transmission mechanisms may be different from standard models.

Interpretation: Policymakers should not infer from expert forecasts alone that the broader economy's expectations are well anchored.

8.3 Low inflation and low attention

- A stable inflation environment may reduce the private value of tracking inflation.
- This can make expectations less responsive to the central bank's target unless information is actively communicated.
- In higher-inflation countries, firms may pay more attention to inflation because it directly affects profits and decisions.

Interpretation: The success of inflation targeting may have an unintended informational side effect: firms may stop paying attention because inflation seems unimportant most of the time.

9 Short summary

- The paper provides rare quantitative evidence on firm managers' macroeconomic beliefs.
- Firms are highly dispersed in expectations, especially for inflation.
- Much of this dispersion reflects misinformation about recent macroeconomic conditions.
- Firms' attention varies with incentives: competition, upcoming pricing decisions, profit slopes, and perceived business relevance.
- Firms update in a Bayesian-like way when given new information.
- Information about the central bank target moves beliefs on impact, but some effects fade over time.
- Belief changes can affect planned investment and employment.
- Inflation is often not perceived as central to business decisions, so firms devote limited resources to tracking it.

10 Conclusion

10.1 Core conclusion

Coibion, Gorodnichenko, and Kumar show that firm expectations are dispersed, imperfectly informed, and strongly shaped by attention incentives. Firms do not simply hold professional-forecaster-like beliefs. They often make large errors about recent inflation, and those errors carry into future inflation expectations.

10.2 Economic intuition

Information is costly. Managers therefore do not monitor every macro variable equally. They pay more attention to variables that are central to their decisions, variables that matter for pricing, and variables that are useful in competitive markets. Aggregate inflation receives limited attention when it is perceived as relatively unimportant.

10.3 Incremental contribution revisited

The paper's contribution is not just documenting disagreement. It connects disagreement to backcast errors, backcast errors to firm incentives, information treatments to Bayesian updating, and belief changes to firm actions. This makes it one of the most complete empirical studies of firm expectation formation.

10.4 Implications for monetary economics

Models in which firms have full information about inflation are incomplete. If managers do not know recent inflation and do not track it closely, then monetary policy through expectations may be weaker, slower, or more heterogeneous than in representative-agent full-information models.

10.5 Limitations

- The evidence comes from New Zealand, a small open economy with low and stable inflation.
- The survey captures general managers, but within-firm decision processes may include other actors.
- Planned decisions are not always identical to realized decisions.
- The experiments reveal responses to supplied information but not the full effect of long-run central bank communication strategies.

10.6 Takeaway

The paper's central lesson is that expectations are economic choices. Firms form beliefs using costly attention, and the allocation of attention depends on the private value of information. Inflation expectations can move when information is provided, but the challenge for policy is getting inattentive firms to notice, believe, and retain the information.